EARLY WARNING FOR SEVERE AIR-POLLUTION DAYS IN DELHI-NCR: PREDICTING RISK AND IDENTIFYING WEATHER AND CALENDAR TRIGGERS TO SUPPORT BUSINESS PLANNING

A Project Report submitted for the course

23CSE452 BUSINESS ANALYTICS

Final Project Report (Units 1, 2 and 3)

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in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Under the guidance of

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NOVEMBER 2026

Declaration

We, the undersigned, hereby declare that the project report entitled **"Early Warning for Severe Air-Pollution Days in Delhi-NCR: Predicting Risk and Identifying Weather and Calendar Triggers to Support Business Planning"**, submitted for the course 23CSE452 Business Analytics at Amrita School of Computing, Amrita Vishwa Vidyapeetham, Coimbatore, is a record of original work done by us under the guidance of Rohini S, Assistant Professor, Department of Computer Science and Engineering. The datasets were collected by the team from the OpenAQ v3 and Open-Meteo archive APIs and a team-built calendar; no pre-built dataset was used. This work has not been submitted, in part or in full, for any other course, degree or diploma. All sources, software and prior studies used are cited, and the use of generative-AI tools is disclosed in Section 8.3.

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Acknowledgement

We would like to express our sincere gratitude to Amrita Vishwa Vidyapeetham and the Amrita School of Computing for providing us with the opportunity and resources to complete this project. We are especially thankful to our project guide, Ms. Rohini S, Assistant Professor, Department of Computer Science and Engineering, for her valuable guidance, support and encouragement throughout the project. We also extend our thanks to the faculty members and everyone who contributed to the successful completion of this work.

Group 11

Abstract

Severe air-pollution days in Delhi-NCR trigger GRAP restrictions that halt construction, slow logistics and cancel school and outdoor events with little notice. We built an early-warning workflow that predicts, up to three days ahead, whether a day will be "very poor or severe" (PM2.5 ≥ 120.5 µg/m³, AQI > 300) and which weather and calendar conditions precede it. The team collected OpenAQ data for 35 stations, hourly Open-Meteo weather and a Diwali calendar, producing 3,195 daily records × 31 variables (2018 – Sep 2026; 792 high-risk days). Gradient boosting predicts next-day risk with recall 0.922 and F1 0.877 on a held-out 2025–26 test set; SARIMAX/gradient-boosting forecasts cut 1/2/3-day MAE from 18.8/24.7/27.5 to 15.7/19.5/19.6 µg/m³ against persistence; FP-Growth yields 113 statistically significant rules, all with test lift > 1. A six-page Streamlit dashboard delivers the alert, an adjustable threshold and a sector playbook. **Keywords:** PM2.5, early warning, SARIMAX, gradient boosting, association rules, Delhi.

Executive Summary

Problem. Construction, logistics, schools and event organisers learn of severe-pollution days too late to reschedule; official AQI describes today and GRAP orders target government action. **Solution.** A 1–3-day PM2.5 forecast and alert, explained by plain if–then rules and tunable to each firm's cost of a missed day, published as the *Delhi Air-Risk Planner* dashboard. **Evidence (Jan 2025 – Sep 2026, 635 days, 128 high-risk):**

- The alert catches about **91% of high-risk days** at every horizon, with 19–23 false alarms in 21 months.
- Forecast error is **16–29% lower** than "tomorrow = today", and the gain grows with lead time.
- **Post-monsoon + low wind + no rain + high pressure** was followed by a high-risk day 93% of the time (29% on average).

Value and actions. If a missed day costs three false alarms, the 3-day alert lowers total cost by 20–44% versus planning on today's air. Adopt a GREEN/YELLOW/RED sector playbook, set the alert near 110 µg/m³ for that cost ratio, confirm with official CPCB/CAQM notices, and add a live data feed before roll-out.

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Abbreviations

Table 1. Abbreviations used in this report

Term	Meaning	Term	Meaning
AQI	Air Quality Index (CPCB)	GRAP	Graded Response Action Plan
ARM	Association Rule Mining	HistGB / GB	(Histogram) Gradient Boosting
BH-FDR	Benjamini–Hochberg false discovery rate	MAE / RMSE	Mean absolute / root-mean-square error
CAQM	Commission for Air Quality Management	PM2.5 / PM10	Particulate matter ≤ 2.5 / 10 µm
CPCB	Central Pollution Control Board	SARIMAX	Seasonal ARIMA with exogenous inputs
DPDP	Digital Personal Data Protection Act, 2023	IMD	India Meteorological Department

1. Introduction, Business Objectives and Literature Review

1.1 Context and scope

When Delhi's AQI enters the Very Poor (301–400) or Severe (401+) band, CAQM invokes GRAP stages that restrict construction, some vehicles and diesel generators and may close schools [1, 2]. **Stakeholders** are construction firms, logistics operators, schools and event organisers. **Objective:** an advance, explainable, tunable signal of whether a coming day will be high-risk (PM2.5 ≥ 120.5 µg/m³). **Scope:** Delhi stations within 25 km of 28.61°N, 77.21°E; daily; 2018 – Sep 2026; PM2.5 only. The alert is a planning aid and does not declare GRAP stages. **Questions:** Q1 when and under what conditions do high-risk days occur (Unit 1); Q2 can next-day risk beat persistence (Unit 2); Q3 how far ahead can PM2.5 be forecast (Unit 3, M1); Q4 which conditions precede a high-risk day (Unit 3, M2); Q5 which threshold minimises business cost.

Table 2. Business objectives, KPIs, baselines and targets

Objective	KPI	Baseline → target	Unit
O1 Detect high-risk days early	Recall (%)	Persistence 87.5 / 82.0 / 79.7 (days 1/2/3) → ≥ 90	U3 M1, U2
O2 Accurate PM2.5 outlook	MAE (µg/m³)	18.8 / 24.7 / 27.5 → lower at every horizon	U3 M1
O3 Limit false alarms	Precision	0.87 (day 1) → ≥ 0.80	U2, U3 M1
O4 Lead time	Days of warning	Same-day AQI → 3 days	U3 M1
O5 Explainable triggers	Validated rules (FDR, test lift > 1)	None → ≥ 20	U3 M2
O6 Usable tool	Deployed app with threshold/cost controls	None → deployed	App

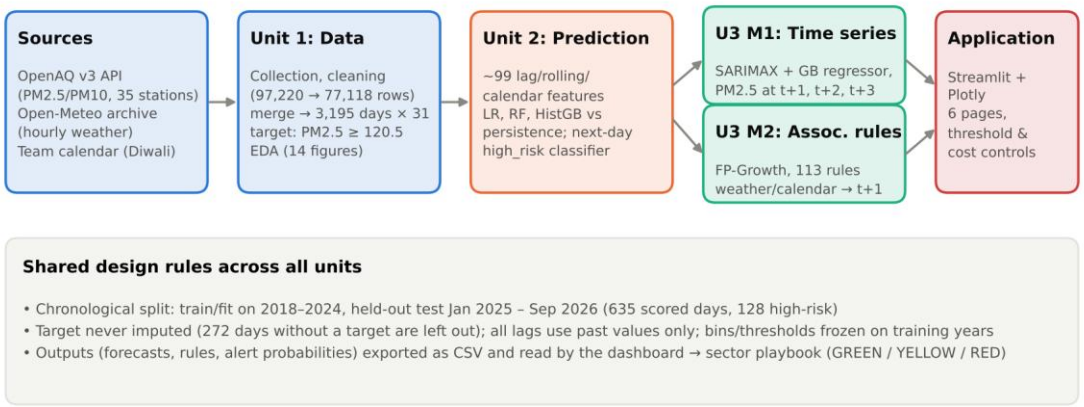


Figure 1. End-to-end workflow linking data collection, Units 1–3 and the dashboard (Matplotlib; drawn for this report).

1.2 Related work and research gap

The **CPCB AQI** [1] reports current conditions; **CAQM GRAP** [2] uses official forecasts to set region-wide enforcement stages. Abirami and Chitra [3] forecast regional air quality with a spatiotemporal deep network on multi-station data; Pande et al. [4] forecast Delhi's daily AQI with LSTM, Bi-LSTM, Bi-RNN and kernel ridge regression (best Bi-RNN: R^2 0.954, RMSE 25.76 AQI units). **Gap:** prior work scores continuous error, not detection of AQI > 300 days; omits the strong persistence baseline; gives no interpretable triggers; and uses fixed thresholds. We score forecasts and alerts against persistence, mine validated rules and expose a cost-based threshold (comparison in Section 6.2).

2. Dataset Collection, Validation and Preparation

2.1 Provenance, coverage and challenges

All data were **collected by the team** on 2 Oct 2026 (`Dataset_Generation.ipynb`); no pre-built dataset was used.

Table 3. Data sources

Source	Method	Raw volume
OpenAQ v3 API [5]	Locations within 25 km (96 found, 35 active); daily values for 137 PM2.5/PM10 sensors; key in Colab Secrets; rate-limit back-off	97,220 sensor-days
Open-Meteo archive [6]	Hourly temperature, humidity, dew point, wind, pressure, rain, cloud at central Delhi → daily	76,680 hours
Team calendar	Month, weekday, IMD season, days from Diwali, ±5-day window	3,195 days

Final dataset: `cleaned_final.csv`, **3,195 days × 31 variables** (1 Jan 2018 – 30 Sep 2026). The target `pm25_target` comes from the only station reporting every year (OpenAQ 8118) on 2,836 days, with a ≥ 10-station city mean on 87 days; 272 days have no target and are never imputed. **2,923 labelled days, 792 high-risk (27.1%).** Train 2018–2024; test Jan 2025 – Sep 2026. **Challenges:** API rate limits (header-driven retries); a monitoring network that grew from 1 to ~30 stations (solved with the reference station, Pearson 0.978 with other stations); thin OpenAQ coverage Nov 2022 – Jan 2025 and no PM10 in 2023–24 (target uses PM2.5 only); and wrong Diwali distances on 272 edge rows (recomputed).

Table 4. Sample records (selected columns)

date	pm25_target	high_risk	temp (°C)	wind (m/s)	RH (%)	rain (mm)	season
2018-11-17	125.0	1	18.5	2.88	63.7	0.0	post_monsoon
2020-11-09	549.0	1	21.0	1.96	52.3	0.0	post_monsoon
2024-01-10	189.0	1	9.2	3.59	84.5	0.0	winter
2026-06-28	53.9	0	36.7	2.23	43.1	1.0	monsoon

Table 5. Core data dictionary and unit mapping

Variable(s)	Meaning / preparation	Used in
pm25_target, high_risk	Daily PM2.5 (µg/m³), reference station with fallback; label = 1 if ≥ 120.5	U1; U2/U3 targets
pm25_mean, pm10_mean	Delhi station means after cleaning	U1; U2 lags
temp, humidity, wind, pressure, precip, cloud	Open-Meteo hourly → daily mean/sum	U1, U2, U3
season, month, days_from_diwali, is_festival_window	Calendar features (IMD seasons)	U1, U2, U3

2.2 Data quality, privacy and ethics

Every issue was counted before change and logged in `M2_cleaning_log.csv`. Weather and calendar files had no gaps, duplicates or range violations; 23 post-merge assertions passed.

Table 6. OpenAQ cleaning rules (97,220 → 77,118 rows)

Rule	Removed
Missing, placeholder (±9999) or negative daily mean	134
< 16 of 24 hourly readings (CPCB rule [1])	19,416
Stuck sensors (min = max) and analyser ceiling codes	541
Single-station spike > 8× peer median	11
Plausible extremes, PM2.5 > PM10, duplicates	flagged / none

Imbalance: 28.9% high-risk in training, 20.2% in test, over 80% in Nov–Jan; handled with class weights and recall-focused metrics. **Privacy (DPDP Act 2023 [15]):** environmental data only, no personal data; API key kept out of the repository; CC BY 4.0 sources. **Bias:** one station represents the city (97.4% label agreement with the city mean); PM2.5-only AQI ignores 22 PM10-driven dust days; a fallback sensitivity test moves the high-risk share by 0.1 points.

2.3 Transformations

Sensor-days → station-days → Delhi city-days (station counts kept), then **left-joined** onto the complete weather–calendar day list so missing pollution stays blank. Imputation, scaling, encoding, log transforms and tercile bins are fitted on training years only, after the chronological split.

3. Exploratory Data Analysis and Visualization (Unit 1)

PM2.5 is right-skewed (mean 95.6, median 67.8, max 549 µg/m³), so medians are used; the 130 Tukey outliers are real winter episodes and kept. Figure 2 shows the same cycle every year: Nov 85%, Dec 86% and Jan 83% of days are high-risk, Oct (26%) and Feb (39%) are

transitions, and Jul–Sep have none (Figure 3). Every winter from 2018–19 to 2025–26 had 55–75% high-risk days, so the problem recurs annually; weekday effects are under 4%.

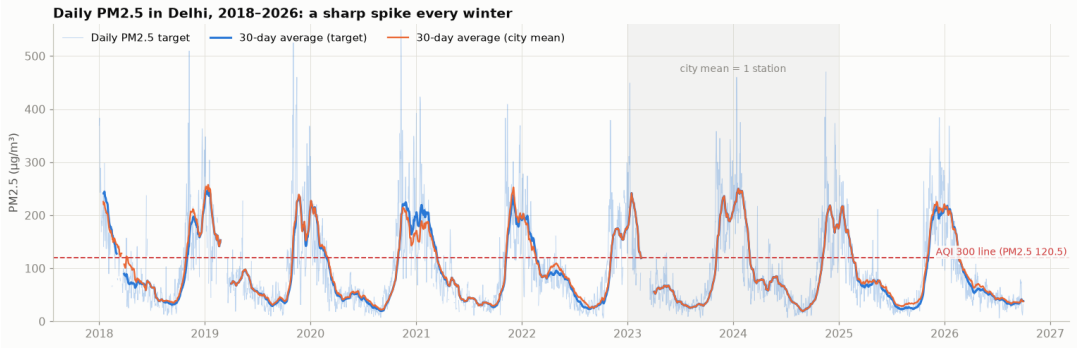


Figure 2. Daily PM2.5, 30-day averages and the AQI-300 line, 2018–2026. Source: M3_EDA.ipynb, cell 17.

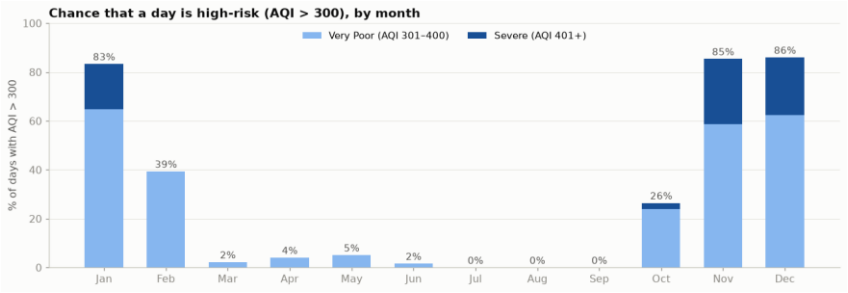


Figure 3. Share of days above AQI 300 by month (Very Poor and Severe). Source: M3_EDA.ipynb, cell 22.

All-year correlations with pressure (0.73), dew point (−0.72) and temperature (−0.59) mostly encode season; within Nov–Jan only wind and gusts (−0.32) remain. Figure 4 shows winter low-wind days are high-risk 94% of the time versus 76% when windy, and rain lowers risk from 86% to 69%. Median PM2.5 rises 74% from the week before Diwali to the five days after, and tomorrow keeps today's high/not-high state on 92.7% of day pairs. **Carried into modelling:** persistence is a hard baseline; season, wind, rain and Diwali are the key non-persistence signals; temperature, pressure and dew point overlap; grouped conditions suit association rules.

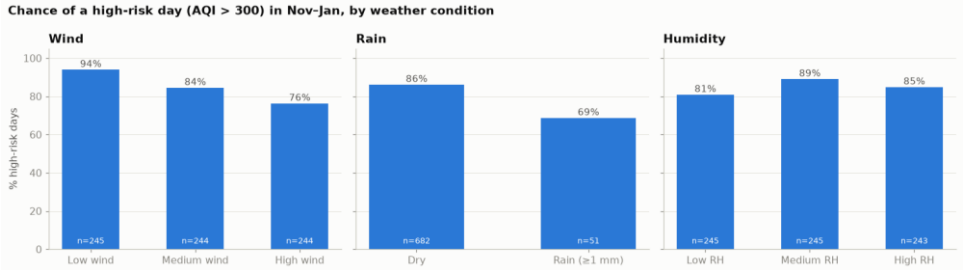


Figure 4. Chance of a high-risk day in Nov–Jan by wind, rain and humidity tercile. Source: M3_EDA.ipynb, cell 35.

4. Feature Engineering and Predictive Modelling (Unit 2)

Features (99 columns, 8 families): lags t−1 to t−3 of pollution and weather; 3/7-day rolling means and deviations; day-over-day changes; weather interactions; cyclical month/day-of-year; winter and Diwali-proximity flags; persistence counts; season dummies. **Protocol:** chronological split (train 2,285 rows to 2024; test 635 rows from 2025); logistic regression, random forest and histogram gradient boosting tuned by 5-fold TimeSeriesSplit on F1, imputation inside pipelines; baseline = persistence. **Leakage audit:** all lags use `shift(1)`; four columns that exactly rebuild same-day PM2.5 were removed (95 inputs remain); same-day weather assumes a forecast is available (tested in 5.1).

Table 7. Model comparison on the held-out test set (635 days, 128 high-risk)

Model	Test metrics	Notes
Persistence	P 0.867, R 0.867, F1 0.867; 17 FA, 17 missed	Warns on 0/16 episode starts
Logistic regression (C = 0.1)	P 0.804, R 0.930, F1 0.862	CV F1 0.861
Random forest (depth 8)	P 0.779, R 0.938, F1 0.851	11/16 episode starts
Gradient boosting (selected)	Acc 0.948, P 0.837, R 0.922, F1 0.877; 23 FA, 10 missed	Best CV F1; winter recall 0.96, Oct/Feb 0.75

Gradient boosting cuts missed days from 17 to 10 for six extra false alarms and, unlike persistence, warns on 8–11 of 16 episode starts. Permutation importance (Figure 5) ranks previous-day AQI far ahead, then day-of-year, previous-day rain, PM2.5 lags, humidity, Diwali distance and wind, matching EDA. It is weakest in October/February transitions (recall 0.75, precision 0.50), exactly when warnings matter, so Section 6 pairs it with forecasts and rules and lets users lower the threshold in those months.

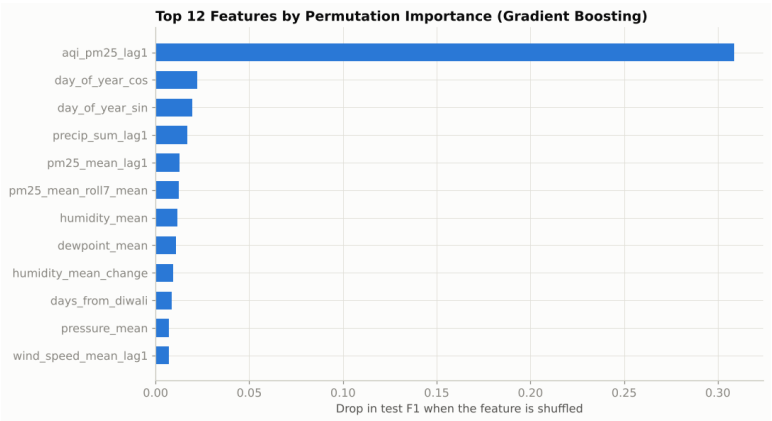


Figure 5. Top 12 features by permutation importance (drop in test F1), gradient boosting. Source: M5_Model_Evaluation.ipynb, cell 18.

5. Advanced Analytics (Unit 3): Two Selected Methods

5.1 Method 1 — Time-series forecasting

Question: how many days ahead can a high-pollution day be anticipated? **Input:** daily `pm25_target` with weather and calendar terms. **Preparation:** log PM2.5, lags 1–7, gaps ≤ 2 days carried forward with flags, targets never filled, weather only as observed at the forecast origin. **Why:** strong autocorrelation and annual seasonality. **Models:** SARIMAX [8] with ARMA(2,0,1) errors and six annual Fourier pairs; a gradient-boosting regressor [7]; four baselines. GB replaces SARIMAX only if it wins 2 of 3 validation folds by > 0.25 µg/m³ (selected: GB day 1, SARIMAX days 2–3). Alert = forecast ≥ 120.5.

Table 8. Method 1 test results (635 days per horizon, 128 high-risk)

Approach	MAE day 1/2/3	RMSE day 1/2/3	Recall day 1/2/3	Precision day 1/2/3
Persistence	18.8 / 24.7 / 27.5	30.7 / 40.0 / 43.5	0.88 / 0.82 / 0.80	0.87 / 0.81 / 0.79
Damped-anomaly persistence	16.4 / 19.8 / 20.6	27.2 / 32.0 / 33.0	0.92 / 0.93 / 0.92	0.86 / 0.86 / 0.83
Selected model	15.7 / 19.5 / 19.6	25.6 / 31.7 / 32.1	0.91 / 0.91 / 0.91	0.86 / 0.83 / 0.83
GB with perfect future weather	15.6 / 18.7 / 19.6	24.8 / 29.5 / 30.6	0.93 / 0.93 / 0.94	0.86 / 0.82 / 0.77

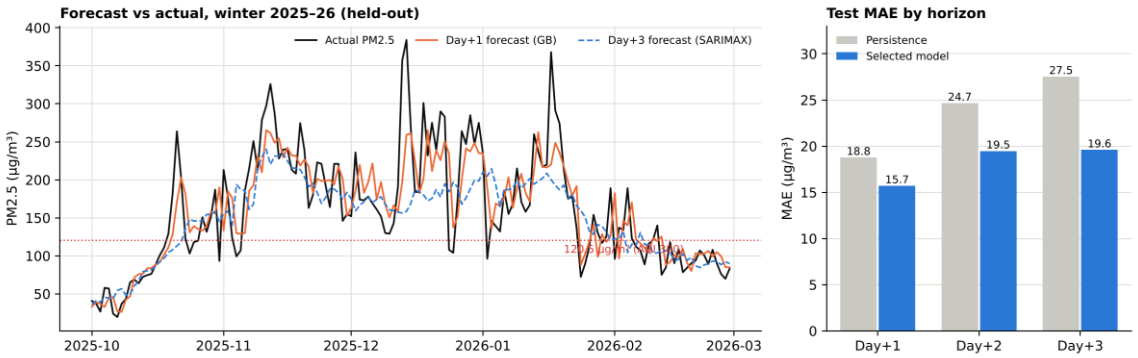


Figure 6. Actual vs day-1 and day-3 forecasts, winter 2025–26, and test MAE by horizon. Source: m3_test_predictions.csv, m3_test_metrics.csv.

MAE falls 16%, 21% and 29% versus persistence for days 1–3 (day-2/3 intervals do not overlap), and the alert catches 116–117 of 128 high-risk days. Forecasts follow episodes but smooth sharp spikes (Figure 6); perfect weather would improve day-1 MAE by only 0.1. **Limitations:** small gain over the damped baseline; in Nov–Jan 88% of days are high-risk; one station; under two test winters.

5.2 Method 2 — Association rule mining (FP-Growth)

Question: which conditions today signal a high-risk day tomorrow? **Input:** 2,792 clean day pairs (t → t+1); 633 frozen 2025–26 test pairs (base rate 19.9%). **Preparation:** wind, humidity, temperature and pressure terciles fitted on training data only (e.g. wind 2.18 / 2.92 m/s); rain,

season, Diwali, weekend; Set A excludes persistence, Set B adds "yesterday high-risk". **Why:** readable if–then reasons with measurable support, confidence and lift. **Settings:** FP-Growth [9, 10], support ≥ 5%, confidence ≥ 0.60, lift ≥ 1.20, consequent high_risk(t+1); redundant rules pruned; BH-FDR at 0.05 [11]; 7-day block bootstrap for lift CIs [12].

Table 9. Method 2: top rules (train vs frozen test)

Rule → high-risk tomorrow	Train supp / conf / lift	Test conf / lift (n)
A: High pressure + Low wind + Mid humidity + No rain + Post-monsoon	5.8% / 93.3% / 3.17	100% / 5.02 (14)
A: Cold + Low wind + No rain + Post-monsoon	6.9% / 93.1% / 3.16	100% / 5.02 (46)
A: Cold + High pressure + Low wind + No rain	9.5% / 90.7% / 3.08	80.4% / 4.04 (107)
B: Cold + Low wind + No rain + Yesterday high-risk	9.8% / 97.7% / 3.32	93.4% / 4.69 (91)

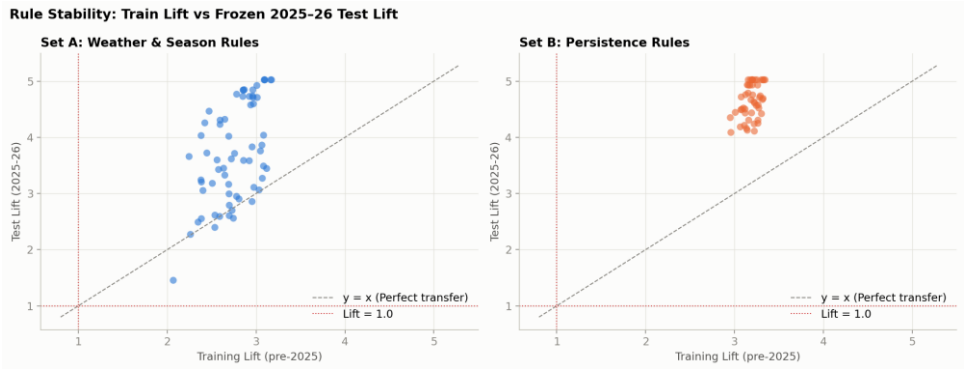


Figure 7. Training vs frozen 2025–26 test lift for Set A and Set B rules. Source: m4 branch, 01_train_vs_test_lift.png.

All **113 rules** (64 Set A, 49 Set B) pass BH-FDR and keep test lift > 1 (Figure 7); 109 are stable and 4 short rules fall below 50% test confidence (the presentation's stricter drift test flagged 28). The common pattern is **stagnation** – low wind, no rain, high pressure, cold, post-monsoon – and weather adds 10–11 points of confidence beyond persistence. **Limitations:** association not causation, some rules rest on 13–14 test days, all-season bins, and rare events like Diwali appear only at 2% support.

6. Comparative Analysis and Business Impact

6.1 Comparison of the two methods

Table 10. Method 1 vs Method 2

Criterion	M1: time-series forecasting	M2: association rules
Output	PM2.5 forecast + alert, days 1–3	113 ranked if–then rules, 1 day ahead
Validation	Held-out MAE/RMSE and alert recall vs 4 baselines	FDR significance, bootstrap CI, frozen test lift
Strength	Quantified error; 3-day lead; 16–29% lower MAE	Plain reasons; shows weather adds value beyond persistence

Criterion	M1: time-series forecasting	M2: association rules
Limitation	Small gain over damped baseline; smooths spikes	Associational; coarse bins; rare events filtered

The metrics are not interchangeable (forecast error vs conditional frequency); the methods are **complementary** – the forecast says *how bad and when*, the rule says *why*. All units agree that calm, dry, post-monsoon days are riskiest and that persistence explains most skill, so value lies at episode starts, longer horizons and Oct/Feb transitions.

6.2 Comparison with existing studies

Table 11. This project vs existing studies

Criterion	Abirami & Chitra 2021 [3]	Pande et al. 2024 [4]	This project
Data / target	Multi-station regional; continuous	Delhi daily AQI; continuous	Delhi PM2.5 + weather + calendar; continuous and AQI > 300 event
Methods / validation	Spatiotemporal deep network; error metrics	LSTM, Bi-RNN, KRR; R^2, RMSE	SARIMAX, GB, LR, RF, FP-Growth; time-series CV, 4 baselines, FDR
Result	Deep models beat statistical baselines	Bi-RNN R^2 0.954, RMSE 25.76 (AQI units)	Day-1 RMSE 25.6 µg/m³; recall 0.91–0.92; 113 rules
Business use	Not reported	No persistence or event recall	Cost-tunable alert; single station

Direct comparison is invalid: Pande et al. report RMSE in AQI units on another period, ours is µg/m³ of PM2.5 on 2025–26, and neither study reports persistence. Our contribution is event-level evaluation against persistence, validated explanations and cost-based thresholds; we make no claim of superiority over deep spatiotemporal models.

6.3 Business impact and scenarios

Costs are in false-alarm units (one unnecessary precaution day); a missed high-risk day costs *r* units: base r = 3, best r = 1, worst r = 5 (assumptions). Counts are measured on the 21-month test period; rupee values are hypothetical.

Table 12. Total cost (false alarms + r × missed), alert at 120.5 vs persistence

Horizon	Persist. FA/missed	Model FA/missed	r = 1	r = 3 (base)	r = 5
Day +1	17 / 16	19 / 11	33 → 30 (−9%)	65 → 52 (−20%)	97 → 74 (−24%)
Day +2	24 / 23	23 / 12	47 → 35 (−26%)	93 → 59 (−37%)	139 → 83 (−40%)
Day +3	27 / 26	23 / 12	53 → 35 (−34%)	105 → 59 (−44%)	157 → 83 (−47%)

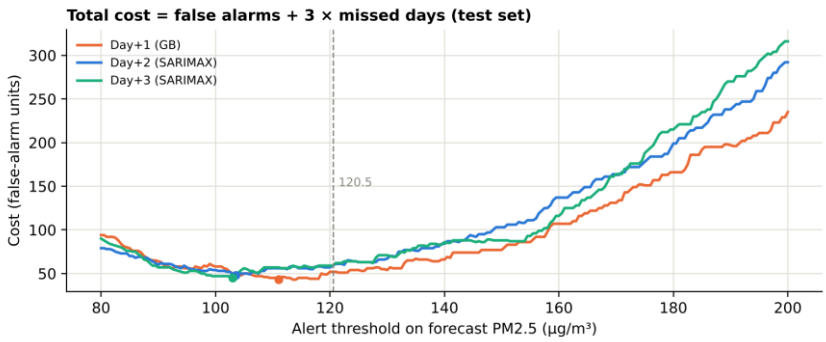


Figure 8. Total cost vs alert threshold under r = 3; dots mark the lowest-cost threshold. Source: m3_test_predictions.csv.

Under r = 3 the cheapest day-1 threshold is about **111 µg/m³** (cost 43 vs 52) and 103–104 for days 2–3 (Figure 8); at 140, false alarms drop to 6 but misses rise to 20. At a hypothetical ₹20,000 per false-alarm unit, the day-3 alert saves about ₹9.2 lakh per site over 21 months. Running costs are near zero (free Streamlit hosting and APIs). Savings depend on the true cost ratio, firms acting on alerts, feed availability and a warm, windy 2025–26 winter.

7. Interactive Application, Figure Index and Reproducibility

Delhi Air-Risk Planner (Streamlit + Plotly [13, 14]): https://delhi-air-risk-planner.streamlit.app/. Notebooks export CSVs to `dashboard/data`; `dash_data.py` loads them; `app.py` renders six pages – Overview (3-day risk cards), Dataset, Forecast (M1), Rules (M2), Insights (threshold, cost ratio, what-if planner) and My business (sector actions). Forecasts and rules are precomputed; alert counts, recall/precision, cost curves and the lowest-cost threshold are computed live. Official CAQM GRAP actions load separately from `grap_actions.csv`; the app never declares a GRAP stage. It replays history rather than running live.

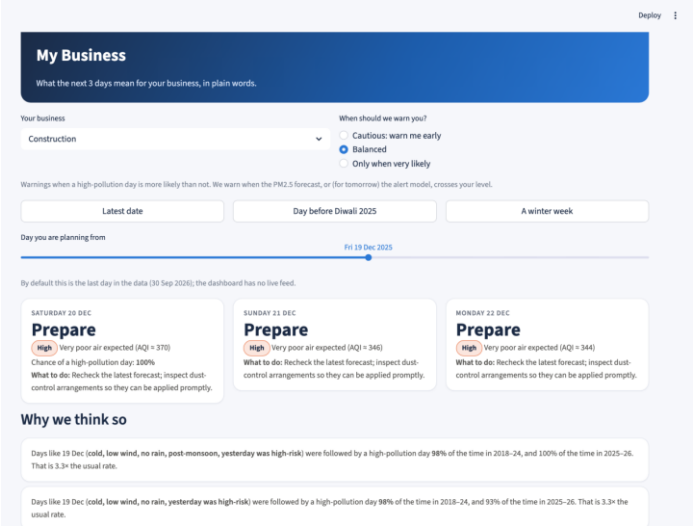


Figure 9. "My business" page: construction outlook from 19 Dec 2025 with the matching rule. Source: deployed dashboard.

Table 13. Figure index

Fig.	Title	Sec.	Source
1	End-to-end workflow	1.1	report script
2–4	PM2.5 trend; monthly risk; winter risk by weather	3	M3_EDA.ipynb cells 17, 22, 35
5	Permutation importance	4	M5_Model_Evaluation.ipynb cell 18
6	Forecast vs actual, MAE by horizon	5.1	m3_test_predictions/metrics.csv
7	Train vs test rule lift	5.2	m4 branch, 01_train_vs_test_lift.png
8	Cost vs alert threshold	6.3	m3_test_predictions.csv
9, A1	Dashboard page; project board	7, App. A	dashboard/app.py; GitHub Projects

Repository: https://github.com/sanjanav2102/Predicting-Risk-and-Identifying-Weather-and-Calendar-Triggers-to-Support-Business-Planning – branches `main` (all notebooks, data, dashboard), `m3-eda`, `m4` (rule mining) and `m5-dashboard`. **Environment:** Python 3.11, pandas, scikit-learn, statsmodels, mlxtend, matplotlib, streamlit, plotly. **Reproduce:** run Dataset_Generation → M2 cleaning → M3_EDA → M4 rules → M5 models (OpenAQ key in `OPENAQ_API_KEY`), then `streamlit run dashboard/app.py`; seed 42.

8. End-to-End Validation, Traceability and Contributions

8.1 Completion and validity

All declared scope is implemented. **Threats to validity:** one reference station and PM2.5-only AQI (internal); Delhi-only, under two test winters (external); a library-version re-run changed classifier F1 from 0.877 to 0.870, and the M1 and feature-engineering notebooks are included in the notebooks/ folder (reproducibility).

Table 14. Objective-to-evidence traceability

Objective	Result	Evidence	Action
O1 recall ≥ 90%	0.91 (forecast); 0.922 (GB)	Tables 7–8, Fig. 6	R1
O2 MAE < persistence	15.7 / 19.5 / 19.6 vs 18.8 / 24.7 / 27.5	Table 8	R1
O3 precision ≥ 0.80	0.83–0.86	Tables 7–8	R2
O4 3-day lead	Forecasts for days 1–3	Fig. 6	R1
O5 ≥ 20 rules	113 significant, all test lift > 1	Table 9, Fig. 7	R3
O6 usable tool	6-page app, live controls	Fig. 9	R4

8.2 Project tracking and individual work

Work was tracked on GitHub Projects (https://github.com/users/sanjanav2102/projects/2; Appendix A) and is backed by dated commits.

Table 15. Team contributions and evidence

Member	Work	Evidence
V Sanjana (CB.SC.U4CSE23751)	Data collection; board; Method 1 time series	Board #5–#9; commits 3 Oct; m3 output files
M Nandan Reddy (CB.SC.U4CSE23431)	Merging and cleaning; GRAP integration	Board #10–#11; upload 3 Oct
Akilan J (CB.SC.U4CSE23002)	EDA; dashboard build and deployment	Board #12; M3_EDA; 8 dashboard commits
Jei Akash M A (CB.SC.U4CSE23518)	Feature engineering; Method 2 rule mining	m4 feature files; m4 branch commit 16 Oct
Isha Sri Prakash (CB.SC.U4CSE23029)	Model data preparation; Unit 2 modelling	M5_Model_Evaluation upload 5 Oct

8.3 AI use and originality declarations

Table 16. Generative-AI use disclosure

Tool	Task supported	Human verification and corrections
Claude (Anthropic)	Compiling all findings and codes to create the live dashboard	Every figure and number was checked by the team against notebook outputs before sign-off
Claude (Anthropic)	Content formatting and image formatting and a draft of the report	Every figure and number was checked by the team against notebook outputs before sign-off

Corrections made during verification: the post-merge assertion count was corrected to 23 from the notebook output; rule drift is reported from the committed rules file (4 rules) alongside the presentation's stricter test (28); Unit 2 metrics use the original run (F1 0.877), with the re-run (0.870) noted in 8.1.

Originality declaration. Data collection, analysis, code and the dashboard are the team's own work. Prior studies, data sources, algorithms and software are cited in the References, and the Amrita logo is used as the university's report insignia. No text, figure or code from other works is used without citation.

9. Conclusion, Recommendations and Roadmap

The project met its objectives: 1–3-day forecasts beat persistence at every horizon, the alert catches about 91% of high-risk days, 113 validated rules explain them, and a deployed dashboard turns them into sector actions. The evidence supports **earlier preparation, not exact prediction**.

Table 17. Prioritised recommendations

#	Recommendation	KPI effect	Owner
R1	Issue a combined 24–72 h alert (forecast level + matching rule) with GREEN/YELLOW/RED sector actions	Recall ≈ 91%; cost −20–44%	Operations heads
R2	Set threshold from own cost ratio (≈ 110 at r = 3; 125–130 at r = 1); lower it in Oct/Feb	Fewer misses in transitions	Risk managers
R3	Show each alert's rule and test confidence; recheck drifting rules each season	Trust	Analytics team
R4	Confirm with official CPCB AQI and CAQM GRAP orders before acting	Compliance	Compliance officers

Roadmap. *0–3 months:* daily live data pull and published sector thresholds. *3–12 months:* multiple reference stations, probability forecasts, cost ratios calibrated with partner firms, success = recall ≥ 90% and precision ≥ 0.80 in winter 2026–27. *> 1 year:* real weather forecasts, other NCR cities and deep spatiotemporal models against current baselines.

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Appendix A: Tracking Board, Commits and Declarations

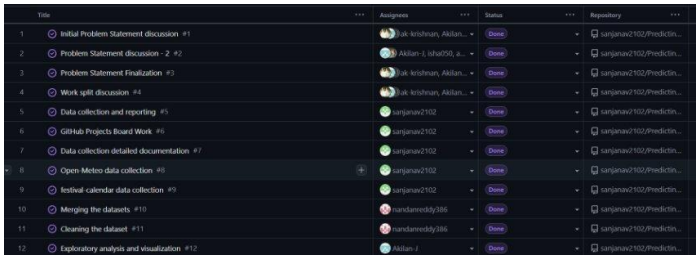


Figure A1. GitHub Projects board (items #1–#12, all Done) with assignees. Source: https://github.com/users/sanjanav2102/projects/2.

Commits (3 Oct – 4 Nov 2026): data files (sanjanav2102), cleaning (Nandan Reddy), EDA (Akilan J), models (Isha Sri), rule mining (ak-krishnan, m4 branch), dashboard (Akilan J, 8 commits). **Dashboard guide:** `cd dashboard`, `pip install -r requirements.txt`, `streamlit run app.py`; required files in `dashboard/data`: cleaned_final, m3_test_predictions, m3_test_metrics, m4_arm_dashboard, m5_test_predictions (+ optional grap_actions). Attach the signed AI declaration and Turnitin report here.